

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	30 September 2026
Team ID	MKU201Team66ab4dee00fc666a751b56cd7
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Creation	Log in to the ServiceNow instance with admin access Create the EEE User (User ID, first name, last name, email)
FR-2	Role Creation	Create the custom roles bb1, bb2, bb3 and bb4 Assign all the roles to the EEE User
FR-3	Table Creation	Create the custom table Institution Details (u_institution_details) Add fields: Student Roll Number, Student Name, Faculty Name, Branch (ECE, EEE, CSE), Email, Phone Number, Description Create records with different branch values (ECE, EEE, CSE)
FR-4	Read ACL (Script-Controlled)	Elevate the security_admin role and create a record-type Read ACL (Advanced) on u_institution_details Require the bb1 role and add the data condition Branch is EEE Add the script that allows admin and bb1 users and denies all others
FR-5	Create ACL	Create a record-type Create ACL on u_institution_details Require the bb2 role with no data condition
FR-6	Write ACL	Create a record-type Write ACL on u_institution_details Require the bb3 role with no data condition
FR-7	Delete ACL	Create a record-type Delete ACL on u_institution_details Require the bb4 role with no data condition
FR-8	Access Verification	Impersonate the EEE User: only EEE branch records are visible A user without the role cannot view any records Impersonate the Admin: all records are visible regardless of branch

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The setup is done through the standard ServiceNow screens (Users, Roles, Tables, Access Control), and users open the Institution Details list as usual and see only the records they are allowed to view.
NFR-2	Security	Record-level access is enforced by Read, Create, Write and Delete ACLs using the roles bb1 to bb4 and a script. Users without the required role are denied, the security_admin role is elevated to create ACLs, and admin keeps full access.
NFR-3	Reliability	The ACL script returns the same allow or deny result for the same role and branch value, so access decisions stay consistent across forms, lists and Service Portal.
NFR-4	Performance	The ACL script only checks roles with gs.hasRole, so access decisions are quick and do not slow down opening the record list.
NFR-5	Availability	The ACLs are active on the ServiceNow instance and are checked every time a record is accessed, with no separate service to keep running.
NFR-6	Scalability	More branches, roles or operations can be supported by adding data conditions, roles or ACLs on the table, and the same rules continue to apply as records grow.